

David Zhao

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EDUCATION

Wilfrid Laurier University

Bachelor of Science in Data Science (co-op) + Big Data Concentration - 3.9 Program GPA

- **Clubs:** Laurier Analytics Society - VP of Tech

Waterloo, ON

Sept 2024 - Present

EXPERIENCE

Software Developer Intern

DOUBL

May 2026 – Present

Toronto, ON, Canada

- **Architected** a split dev/prod Firebase environment with isolated Firestore databases and storage buckets, eliminating test data contamination and protecting **1,000+ real user** records from dev-cycle risk, improving development workflow safety
- **Upgraded** Doubl's static website into a full-stack ML-powered platform by integrating **33+ responsive**, mobile compatible pages with business-driven features, accelerating the product's readiness for early user onboarding
- **Collaborated** with a senior engineer and product owner through daily standups and weekly alignment meetings in an Agile environment, surfacing blockers and aligning on sprint priorities to consistently ship features on schedule

Software Developer Intern

Emergconnect

January 2026 – April 2026

Waterloo, ON, Canada

- **Optimized an NLP classification pipeline**, reducing patient complaint **misclassification rate by 25% across 5 triage categories**, by refactoring model logic and implementing systematic prompt refinement and error analysis
- **Built automated evaluation pipelines** using test case design and confusion matrix analysis, identifying edge cases across 200+ patient samples and refining classification thresholds to improve **prediction consistency by 30%**
- **Documented** and presented findings reports to the project manager detailing model performance metrics and optimization outcomes, directly influencing the team's decision to adopt the refined NLP pipeline

Instructional Assistant

Wilfrid Laurier University

September 2025 – Present

Waterloo, ON, Canada

- **Supported** 60+ students in understanding core data analytics concepts including data cleaning, hypothesis testing, regression, and ANOVA, breaking down complex statistical methods into clear, approachable explanations
- **Debugged** student R code and guided data visualization using ggplot2 and Matplotlib, helping students troubleshoot errors and understand not just the solution but the reasoning behind each method
- **Delivered** structured Python and R coding walkthroughs through office hours and tutorials, adapting explanations to each student's background to build their coding confidence and analytical reasoning

PROJECTS

- 🔗 **GreenLens AI - 1st Place Data Mind Challenge** | *Python, Pandas, Matplotlib, ReportLab* March 2026
 - **Engineered** an AI sustainability analytics platform that processed enterprise token usage data across models and departments, converting raw metrics into cost, carbon, and water impact calculations
 - **Generated** automated PDF reports using **Python and ReportLab**, transforming raw AI usage data into KPI summaries and business recommendations, enabling non-technical stakeholders to make data-driven AI governance decisions
 - **Processed** and cleaned enterprise AI usage data using **Pandas**, calculating cost, carbon, and water impact metrics across model types and departments, transforming raw token activity into targeted optimization recommendations
- 🔗 **A.U.R.A. - Winner of Bearhacks** | *API, Cloud Storage, Three.js, Vultr* April 2026
 - **Engineered** a three-agent AI pipeline (Geometry Extraction, Placement Planning, Scene Assembly) using **Backboard APIs**, transforming 2D floor plans into structured data and fully interactive 3D environments
 - **Architected and deployed** a full-stack application on **Vultr** (VPS, Object Storage, PostgreSQL, Load Balancer), enabling scalable scene generation, persistent storage, and real-time job tracking for end-to-end workflows
 - **Collaborated** in a cross-functional team using GitHub to design intuitive user experiences and align AI outputs with real-world spatial needs, ensuring the product was robust and practically usable for diverse use cases
- 🔗 **CareSync - Winner HackCanada** | *Auth0, JavaScript, Node.js, React* March 2026
 - **Built an AI-powered care coordination platform** by implementing persistent patient memory and automated briefing generation, streamlining communication between PSWs, coordinators, and families
 - **Developed a multi-agent system** where automated agents detect medication discrepancies through nightly monitoring, generate caregiver handoff briefings, and draft personalized family updates from clinical notes
 - **Implemented** secure authentication and role-based access control using **Auth0**, protecting sensitive healthcare data across multiple user types including PSWs, coordinators, and families, while enforcing compliant, least-privilege access

TECHNICAL SKILLS AND CERTIFICATIONS

Achievements: 6X Hackathon winner, Harvard CS50

Languages: C/C++, CSS, HTML, Java, JavaScript/TypeScript, Python, R, SQL (PostgreSQL)

Frameworks & Libraries: Flask, FastAPI, Next.js, Node.js, NumPy, Pandas, React, scikit-learn, Three.js, Vue.js

Tools: AWS, Azure, Copilot, Docker, Firebase, GitHub, Google Cloud, Jupyter Notebook, MongoDB, MySQL, Vercel, VS Code